

ZIMBANK AI CREDIT UNDERWRITING PLATFORM

Production-Grade Lending Decision Engine Architecture & Operations Manual

VERSION: 2.1

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DATE: March 2026

1. EXECUTIVE SUMMARY & CORE ENGINEERING MANDATE

The **ZimBank AI Credit Risk Underwriting Platform** is an enterprise-tier, high-throughput, low-latency financial scoring system custom-engineered for banking institutions, micro-lenders, and retail fintech ecosystems across Zimbabwe and the broader Southern African Development Community (SADC). Operating in volatile macroeconomic frameworks requires scoring methodologies that cross-examine non-linear behavioral correlations without relying strictly on static, outmoded centralized credit bureau indices.

By consolidating multi-model machine learning stack architectures with rigorous human-interpretable feature attribution frameworks, the platform satisfies two fundamentally opposing vectors in fintech engineering: predictive maximization and strict regulatory auditability.

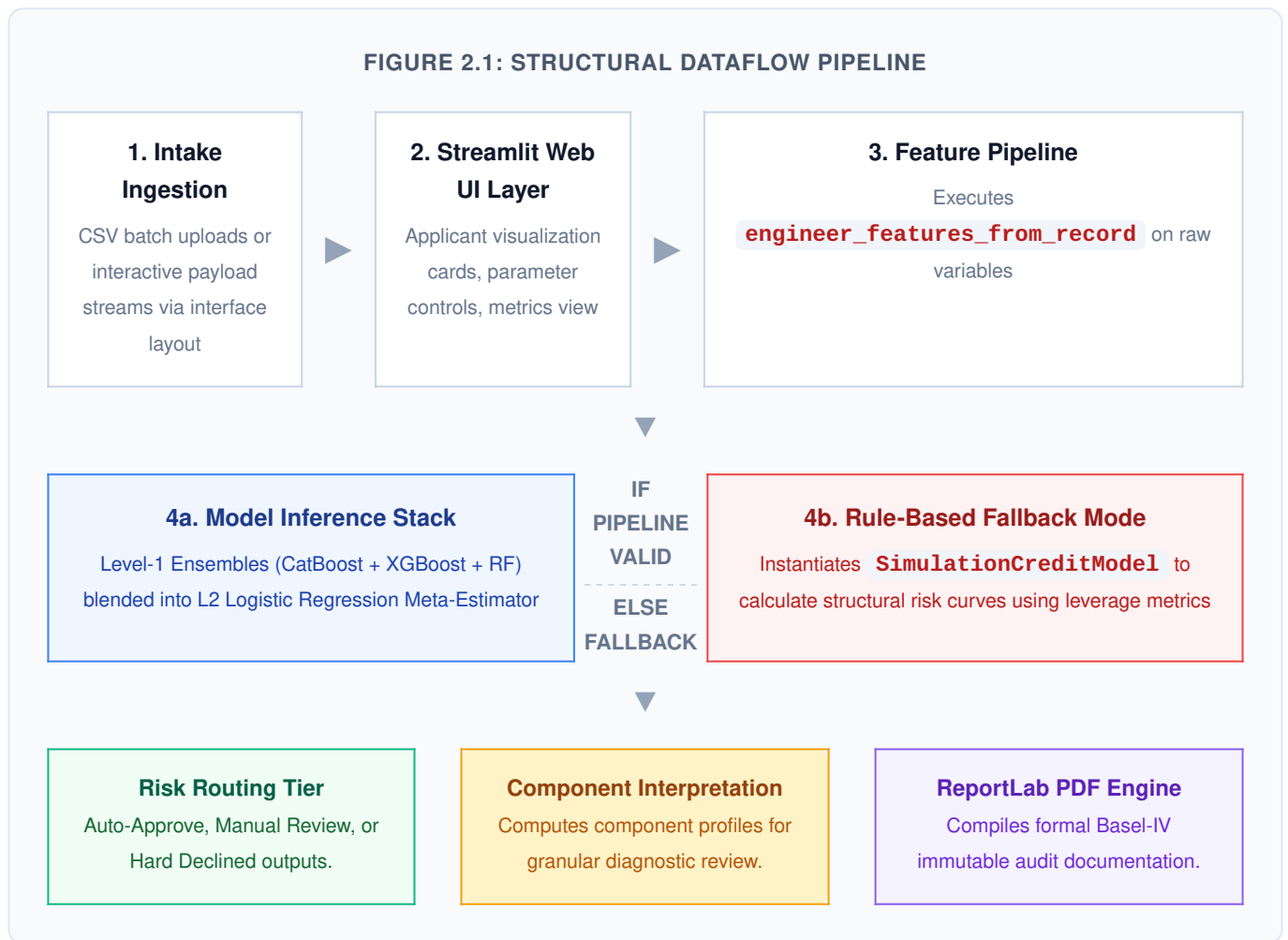
CORE PERFORMANCE & INTEGRATION SLA REQUIREMENTS

- **Automated Inference Latency:** Sub-500ms processing from raw ingestion payload to formal decision assignment.
- **Explainability Threshold:** 100% deterministic traceability for every prediction utilizing component risk factor breakdown metrics.
- **Regulatory Compliance:** Strict technical alignment with Basel IV capital adequacy assessment workflows, providing automated, immutable cryptographic audit reporting.
- **System Fault Invariance:** Layer-0 deterministic fallback execution matrix designed to eliminate core processing failures if predictive artifacts are missing.

2. SYSTEM ARCHITECTURE & PIPELINE DATAFLOW

The infrastructure utilizes a highly decoupled multi-stage logical pipeline where functional modules maintain clear boundary separation. Ingestion layers feed straight into validation structures, which transform inputs before invoking the runtime predictive engine.

FIGURE 2.1: STRUCTURAL DATAFLOW PIPELINE



2.1 Serialization & Pipeline Initialization Execution (`load\_validated\_pipeline`)

System robust initialization relies upon structural verification of binary model representations. The platform boots via an order-of-operations cascade, designed to gracefully drop down in features if underlying artifacts are missing or fail checksum validation:

1. **Primary Pipeline Stack Bundle** (`credit\_risk\_stack.joblib`): A unified data structure map containing individual model objects: `meta_model`, `catboost`, `xgboost`, and `random_forest`.
2. **Standalone Target Model** (`final\_credit\_risk\_model.joblib`): Loaded if the complete stack bundle is unavailable. It checks if the binary is a full meta-model or a direct standalone Scikit-Learn estimator.
3. **Sub-System Deterministic Fallback Mode** (`SimulationCreditModel`): If serialization fails completely due to file corruption, missing links, or infrastructure changes, the platform initializes a calibrated mathematical rule-engine simulator. This guarantees zero workflow downtime in operational retail lending environments.

3. FEATURE ENGINEERING PIPELINE & MATHEMATICAL SPECIFICATIONS

Raw transactional arrays are mapped through programmatic transforms in `engineer_features_from_record`. This process ensures total determinism between the statistical parameters utilized during training and the live scoring matrix engine.

3.1 Formal Transformations & Calculations

TARGET FEATURE KEY	ALGORITHMIC IMPLEMENTATION FORMULA	ECONOMIC CREDIT RATIONALE
monthly_installment	$M = A / (T + 0.1)$ Where A = Principal USD, T = Loan Term Months	Derives gross absolute month-on-month liability. Constant factor prevents division-by-zero errors.
dti_ratio	$DTI = M / I_{monthly}$ Where $I_{monthly}$ = Certified Monthly Net Income	Measures real debt burden relative to capacity. Primary vector for consumer bankruptcy estimation.
total_to_income	$LTIR = A / I_{monthly}$	Establishes macroscopic exposure leverage footprint relative to baseline monthly revenue cashflows.
work_stability	$S = M_{employer} / (Age \times 12)$ Where $M_{employer}$ = Seniority months at current workplace	Normalizes employee tenure across age demographics. Captures employment resilience trends.
has_obligations	$O = [existing_obligations > 0] \rightarrow 1 : 0$	Boolean check flagging co-existing credit relationships or outstanding market debts.

Categorical features (**province** , **employment_sector** , **loan_purpose** , **product_code**) are preserved as raw object strings to leverage native, high-dimensional target-encoding structures embedded directly inside the CatBoost inference engine.

3.2 Granular Risk Component Score Mapping (`calculate_risk_score_components`)

To facilitate human interpretation, the platform breaks down risks into five bounded sub-components scaled from **0** (minimum risk) to **100** (extreme risk). These components form the foundation of the regulatory narrative generation pipeline.

RISK COMPONENT	DETERMINISTIC SCORING LOGIC CONDITIONS	RISK ASSESSMENT CLASSIFICATION
DTI Burden Score	If $DTI < 0.20 \rightarrow 10$ Else if $DTI < 0.35 \rightarrow 25$ Else if $DTI < 0.50 \rightarrow 50$ Else $\rightarrow \min(DTI \times 100, 95)$	$Score < 30 \rightarrow$ Low Burden $30-60 \rightarrow$ Elevated Burden $> 60 \rightarrow$ Critical Risk
Employment Stability	If $M_{employer} < 6 \rightarrow 75$ Else if $M_{employer} < 12 \rightarrow 50$ Else if $M_{employer} < 24 \rightarrow 30$ Else $\rightarrow 10$	Highly nonlinear tracking to capture probationary employment volatility.
Existing Leverage	$\min(existing_obligations \times 20, 80)$	Linear risk multiplier scaling directly with parallel open exposures.
Life Stage Profile Risk	If $Age < 25 \rightarrow 45$ Else if $Age < 35 \rightarrow 25$ Else if $Age < 60 \rightarrow 15$ Else $\rightarrow 35$	Captures demographic variance in earning stability and retirement exposure.
Loan-to-Income Index	If $LTIR < 0.10 \rightarrow 10$ Else if $LTIR < 0.20 \rightarrow 25$ Else if $LTIR < 0.35 \rightarrow 45$ Else $\rightarrow \min(LTIR \times 100, 90)$	Evaluates gross systemic request sizing relative to income reality.

4. HARDCODED BOUNDARY RULES & DECISION CONTROL MATRICES

The runtime execution system applies hardcoded parameters defined at the top of the system configuration file `app.py`. These boundaries are non-modifiable at runtime to guarantee process immutability.

```
# System Decision Boundary Control Matrix
POD_APPROVE_MAX = 0.35 # Cumulative Probability of Default floor for Auto-Approval
POD_REVIEW_MAX  = 0.60 # Probability boundary dividing manual review from auto-decline
DTI_CRITICAL     = 0.42 # Macroeconomic trigger threshold for high-leverage risk routing
```

4.1 Comprehensive Score Routing Logic

CALCULATED POD RANGE	ASSIGNED RISK CLASS	SYSTEM OPERATIONAL ACTION	PDF REPRESENTATION COLOR PROFILE
$POD < 0.35$	Low Risk (Tier A)	✓ AUTO-APPROVED	Institutional Success Green (#10b981)
$0.35 \leq POD < 0.60$	Medium Risk (Tier B)	⌚ CREDIT UNDERWRITER REVIEW	Warning Warning Amber (#f59e0b)
$POD \geq 0.60$	High Risk (Tier C)	✗ SYSTEM HARD DECLINED	System Error Crimson (#ef4444)

4.2 Critical Factor Flagging Systems

Even when the overall model prediction falls within acceptable boundaries, individual structural parameters can trigger automated diagnostic overrides. These factors generate specific system alerts and appear as critical risk drivers in the automated narrative generator:

- **Debt-to-Income Exposure:** Triggered if `dti_ratio > 0.42` . Indicates systemic payment vulnerability under sudden microeconomic changes.
- **Tenure Instability:** Triggered if `months_at_employer < 12` . Flags high probationary employment risk.
- **Credit Market Leverage:** Triggered if `existing_obligations > 2` active accounts. Indicates over-extended consumer profiles.
- **Demographic Burden:** Triggered if `num_dependents > 3` . Flags heightened non-discretionary baseline spending requirements.

5. BASEL IV COMPLIANT PDF GENERATION ARCHITECTURE

The generation of reporting artifacts is handled by the `compile_underwriting_pdf` engine, utilizing ReportLab to output directly into an in-memory `BytesIO` object. This architecture ensures absolute security by avoiding unnecessary disk caching operations.

5.1 Structural PDF Content Architecture

1. **Executive Decision Summary Block:** Displays the transaction fingerprint, unique Application ID tracking token, calibrated Probability of Default (POD), risk classification category, and system-authorized lending action.
2. **Structured Exposure Profile:** Compiles verified absolute dimensions including requested loan size, term structure, current net income, province metrics, and sector parameters.
3. **Granular Component Matrix Table:** Renders multi-tiered sub-component ratings using clear text labels and dark institutional layout tones (`#0f172a` header backgrounds with clean `#cbd5e1` rule borders).
4. **Contextual AI Underwriting Narrative:** Programmatic textual generator that analyzes and constructs natural language summaries of primary risk drivers using conditional text blocks.

5. **Cryptographic Authorization / Audit Trail:** Formally places signature block fields for both the automated machine learning engine validation token and the validating banking officer.

6. USER INTERFACE ARCHITECTURE & LAYOUT FRAMEWORK

The platform's frontend is constructed using an optimization layout inside Streamlit to provide structured workflows for high-volume banking operators.

6.1 Sidebar Operator Controls

- **Applicant Identity Lookup:** Interactive dropdown structure filtering by customer identifier keys. Triggers immediate reactive refreshes across the downstream predictive pipeline.
- **Inference Engine Status Monitor:** Real-time hardware telemetry display flagging the active operational pipeline level (`stacking_full` , `stacking_meta` , `sklearn_direct` , or `simulation_fallback`).

6.2 Main Operational Cockpit Layout

- **Applicant Profile Grid:** Segmented information blocks detailing Financial Capacity, Employment Metrics, and Geographic/Product attributes.
- **Decision Engine Dashboard Header:** A large visual readout box that updates dynamically with the assigned risk classification and decision response.
- **Primary Technical Metrics Row:** High-impact key metrics showing Default Probability, Debt-to-Income (DTI) values, and monthly payment liabilities.
- **Visual Score Component Bars:** Renders inline colored progress bars representing the sub-component risk scores for immediate clinical interpretation.
- **Immutable PDF Exporter Button:** Triggers the underlying ReportLab flow, feeding into `st.download_button` without requiring page-state refreshes.

6.3 Enterprise Batch Processing Pipeline

When operators invoke the `Run Batch Scoring on All Applicants` routing function, the engine changes states to process the entire source dataset. It executes feature transformation and multi-model inference loops across the target matrix, tracking execution with an inline progress widget.

Once calculations complete, the interface renders an interactive results table containing columns for `ID` , `default_probability` , `risk_tier` , and `dti_ratio` , alongside an active exporter module to download raw CSV outputs.

7. OPERATIONS RUNBOOK & DEPLOYMENT PROTOCOLS

7.1 Local Sandbox Environment Provisioning

Execute the following commands in sequence to stand up an isolated instance of the underwriting system:

```
# 1. Clone repository from secure source control
git clone https://github.com/your-org/zimbank-ai-underwriter.git
cd zimbank-ai-underwriter

# 2. Provision isolated virtual environment
python -m venv venv
source venv/bin/activate # Executed for POSIX shells
# venv\Scripts\ctivate # Executed for Windows targets

# 3. Ingest deterministic system requirements
pip install -r requirements.txt

# 4. Initialize production-grade local Streamlit daemon
streamlit run app.py --server.port 8501 --server.address 0.0.0.0
```

7.2 Immutable Dependency Matrix (`requirements.txt`)

```
streamlit>=1.28.0
pandas>=2.0.0
numpy>=1.24.0
joblib>=1.3.0
reportlab>=4.0.0
scikit-learn>=1.3.0
catboost>=1.2.0
xgboost>=2.0.0
weasyprint>=61.0
```

7.3 Enterprise Production Deployment Topologies

For enterprise deployments, standard interactive servers are avoided in favor of immutable container structures or managed processing runtimes:

- **Containerized Docker Topologies:** Bundle the application layer inside minimal Alpine-based base patterns, exposing port **8501** into high-availability clusters such as AWS ECS or Azure Container Instances.
- **Systemd Linux Service Configuration:** For standalone internal server hosting, the process should be managed as an isolated system service to ensure automatic recovery on failure.

8. ENGINEERING TROUBLESHOOTING & SYSTEMS DIAGNOSTICS MATRIX

OBSERVED SYSTEM ERROR TRACE	LIKELY ARCHITECTURAL FAILURE ROOT CAUSE	TARGET ENGINEERING REMEDIATION PROTOCOL
<code>KeyError: "Missing required columns..."</code>	Ingested CSV dataset schema misses key tracking features such as <code>amount_usd</code> , <code>term_months</code> , or <code>monthly_income_usd</code> .	Validate intake ETL schemas against the structural schema mapping definition in Appendix 9.1. Ensure raw keys exactly match lowercase structures.
<code>"SYSTEM WARNING: Model operating in fallback mode"</code>	The system failed to detect or deserialize <code>final_credit_risk_model.joblib</code> or the stack dictionary array artifact.	Verify model binary storage paths. Confirm deployment build steps correctly move down model storage files into the active workspace directory.
<code>ReportLab / WeasyPrint Generation Fault</code>	System font definitions are missing or rendering engine dependencies fail to bind during dynamic library linking.	Verify OS font packages exist (<code>msttcorefonts</code> on Linux). Re-run setup scripts to confirm underlying C-library wrappers link correctly.
Extreme compute slowdown during batch scoring runs	Row-by-row iteration loops create processing bottlenecks when evaluating large data arrays (e.g., portfolios exceeding 10k rows).	Refactor internal scoring loops to run vectorized operations using native Pandas matrix functions instead of single-row iteration methods.

9. SYSTEM TECHNICAL APPENDIX

9.1 Reference Input Record Structure Spec (`test.csv`)

```
ID,amount_usd,annual_rate_pct,term_months,monthly_income_usd,existing_obligations,num_dependen
ts,months_at_employer,province,employment_sector,loan_purpose,product_code,client_dob
APP001,2500,24.5,18,550,1,2,24,Harare,Retail Trade,Business Expansion,PL01,15/03/1995
APP002,12000,19.0,36,2100,3,1,72,Bulawayo,Manufacturing,Asset Acquisition,BL04,22/07/1984
```

9.2 Structural Outline of standard ReportLab / Basel-IV PDF Token Outflow

```
+-----+
| CREDIT UNDERWRITING ASSESSMENT REPORT |
| Basel IV Asset Evaluation Framework • Generated: 18 May 2026 14:32:10 UTC |
```

EXECUTIVE DECISION SUMMARY		
Application ID:	APP001	Default Probability: 12.3%
Risk Classif.:	Low Risk (Tier A)	Underwriting Status: APPROVED
CREDIT EXPOSURE PROFILE & VARIABLE MATRIX		
Principal Amount:	\$2,500 USD	Calculated DTI Ratio: 0.25
Term Structure:	18 Months	Monthly Net Income: \$550 USD
UNDERWRITING RISK COMPONENT NARRATIVE BREAKDOWN		
System calculated a DTI burden index of 25, implying strong capacity.		
Employment tenure parameters indicate high stability within targeted sector.		
SYSTEM AUTHORIZATION & AUDIT TRAIL SIGNATURES		
AI Engine Validation Checksum: [0x8F2D3A9E...]		
Credit Officer Validation Signature: _____		

10. DOCUMENT GOVERNANCE & OPEN SOURCE LICENSE TERMS

- **Operational Distribution License:** MIT Enterprise License – permitted for custom adaptation across corporate environments.
- **Development Track Alignment:** Zimbabwe Fintech Challenge (Specialized Track: Advanced Financial Inclusion Models).
- **Platform Dependencies:** Streamlit Framework, ReportLab Document Suite, Scikit-Learn Ecosystem, CatBoost Architecture, XGBoost Ensemble Engines, Pandas Data Science Suite, NumPy Array Matrix Libraries.

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